INCH-POUND

MIL-T-44237B
30 June 1991
SUPERSEDING
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26 July 1988

MILITARY SPECIFICATION

TURKEY SLICES WITH GRAVY, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers turkey slices with gravy, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-44340	- Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-900	- Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

Food Safety and Inspection Services, Canning of Meat and Poultry Products (9 CFR Part 318)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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U.S. Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-Sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

* 3.2.1 Turkey. Turkey shall be prepared from chilled or frozen ready-to-cook yearling turkeys or young turkeys or a combination of both classes which has been processed in accordance with USDA Poultry Inspection Regulations. Turkey shipped between plants shall be accompanied by a USDA Poultry Products Grading Certificate to certify class and condition of the product and either the initial chilling date or initial freezing (in-storage) date.

* 3.2.1.1 Turkey, chilled. Raw bone-in or boneless turkey shall be received in the chilled state. The turkey shall not have been previously frozen. The chilled turkey shall be held for not more than 6 days after initial chilling at a temperature not to exceed 40°F prior to preparation and further processing (see 3.3).

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* 3.2.1.2 Turkey, frozen. Raw bone-in or boneless turkey shall be received in the frozen state. The turkey shall be held for not more than 120 days after initial freezing at a temperature not to exceed 0°F prior to preparation and further processing (see 3.3).

* 3.2.1.3 Turkey, cooked, frozen. Frozen cooked turkey shall be prepared from chilled turkey that complies with 3.2.1 and 3.2.1.1. Turkey shall be processed in accordance with 3.3.1 and 3.3.1.1. Cooked turkey, in either log or slice form, shall be packaged and sealed in USDA food packaging material and shall be frozen to an internal temperature of 0°F or below within 72 hours. Packaged, frozen, cooked turkey shall have been held at 0°F or below for a period not to exceed 60 days prior to tray pack filling (see 3.4). The initial freezing (in-storage) data and processing and packaging requirements shall be certified by a USDA Agricultural Marketing Service (AMS) Poultry Products Agent.

3.2.2 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.4 Sodium tripolyphosphate. Sodium tripolyphosphate shall comply with the Food Chemicals Codex.

3.2.5 Pepper, white, ground. The ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner covering has been removed. The ground pepper shall have a characteristic penetrating odor, a hot, biting, pungent flavor, and a light color. The ground white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.6 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.7 Broth, chicken. Chicken broth may be frozen or canned (thermostabilized). The chicken broth shall be produced from USDA inspected fowl in accordance with USDA Meat and Poultry Inspection Regulations. It shall be free from extraneous material or cracklings. If frozen, broth shall have been held at an internal temperature of 0°F or below for a period not to exceed 75 days prior to use in product formulation. The broth shall be clear, essentially fat free, and concentrated to a soluble solids level sufficient to comply with the solids requirement in the product formulation. The broth shall have a characteristic mild chicken broth odor and flavor and may contain flavor enhancers approved by FDA.

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3.2.8 Starch, food, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The modified starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked slurry prepared from the starch shall be bland with essentially no cereal or starch taste. The starch shall comply with MIL-STD-900.

3.2.9 Onion powder. Onion powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.10 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.11 Oil, vegetable. The vegetable oil shall have a clean, bland flavor; shall have a minimum stability of 15 hours (A.O.M), a free fatty acid value not to exceed 0.05 percent by weight, and a moisture/volatile matter not to exceed 0.06 percent by weight; and shall pass a cold test of 5.5 hours (minimum).

3.2.12 Celery seed, ground. Ground celery seed shall be prepared from the seed of Apium graveolens L. The ground celery seed shall be light to rich brown in color and possess a characteristic celery odor and flavor with a warm, slightly bitter taste. The celery seed shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 35 sieve.

3.2.13 Sage, ground. Ground sage shall be the dried leaves of Salvia officinalis L. The ground sage shall be green to grayish-green in color; shall possess a strong, fragrant, aromatic odor, and a slightly bitter taste. The ground sage shall contain not less than 1.5 mL of volatile oil per 100 grams of ground sage and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.14 Turmeric, ground. Ground turmeric shall be prepared from Curcuma longa L. and shall possess a deep yellow to orange-yellow color with a peppery, aromatic odor. The ground turmeric shall contain not less than 3.5 mL of volatile oil per 100 grams of ground turmeric and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.15 Marjoram, sweet, ground. Ground sweet marjoram shall be derived from the dried leaves of Majorana hortensis M. The ground marjoram shall be light green to light gray-green in color and possess an aromatic, slightly sharp flavor. The ground marjoram shall contain not less than 0.6 mL of volatile oil per 100 grams of ground marjoram and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

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3.2.16 Flavoring, thyme. Soluble thyme flavoring shall be prepared from extracts of dried leaves of Thymus vulgaris L. by blending and dispersing a minimum of 4.0 percent of total extractives of thyme on a soluble, dry, edible carrier.

3.2.17 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Turkey preparation. Turkey shall be made boneless and skinless. The wing tips, giblets, neck, backs, feet, and viscera shall be excluded. The boned meat shall be free from pin feathers, skin, blood clots, bruises, blemishes, ligaments, tendons, coarse connective tissue, cartilage, and bone measuring more than 0.3 inch in any dimension. The boneless meat shall be comprised by weight of at least 50 percent white meat. Cooked turkey slices shall be prepared in accordance with 3.3.1.1.

* 3.3.1.1 Turkey log preparation and processing. The turkey log shall be prepared and processed as follows:

Ingredient	Percent by weight
Turkey meat	96.50
Ice or ice water	2.19
Salt	1.00
Sodium tripolyphosphate	0.25
Pepper, white, ground	0.04
Garlic powder	0.02

a. Fresh-chilled turkey meat processed on the day of boning shall be coarse ground within 24 hours following boning. If fresh-chilled (not previously frozen) turkey meat is transported between plants, it shall be maintained at an internal temperature of 28° to 40°F and coarse ground within 6 days following initial chilling. Frozen turkey meat shall be maintained at 0°F or lower until further processing and be coarse ground within 120 days after boning. The meat shall be mechanically coarse ground once through a kidney-shaped plate, having openings measuring 2 inches in the least dimensions, and using a two-bladed knife that yields chunked pieces.

b. The chunked turkey pieces shall be mechanically mixed with the spices and other ingredients. The mixing shall continue until the mixture exhibits a sticky (tacky) consistency. Time from grinding to mixing shall not exceed 4 hours. The ground turkey meat shall be maintained at an internal temperature of 28° to 40°F during this time period.

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c. The mixed meat shall immediately be tightly stuffed into casings of a type and size to accommodate the cooking method and the finished product meat dimension requirement. Alternatively, the mixed meat may be formed in stainless steel molds to a size which will ensure compliance with finished product meat dimension requirements. The formed meat logs shall be in the cooking process within 4 hours after being formed. (This period of time may be extended for up to 24 hours provided the meat logs are maintained at an internal temperature of 28° to 40°F.)

d. The turkey logs shall be cooked in a cook house (smoke house without smoke) or by other commercially acceptable cooking methods to provide a product meeting the finished product drained weight and other requirements. Water cooking is permitted only for the meat logs in casings provided they are cooked in water until the internal temperature of the log reaches 150° to 152°F.

e. Immediately after completion of the cooking process, the cooked meat logs shall be placed in an ice water bath, or cooled by other commercially acceptable cooling methods, and rapidly cooled to 130°F or below. The turkey logs shall be further cooled from 130° to 80°F or below within 2 hours, and further cooled from 80° to 40°F or below within 4 hours. The cooled, cooked logs shall be held for not more than 48 hours at an internal temperature of 28° to 40°F after removal from the cooking process and prior to slicing or packaging for frozen storage (see 3.2.1.3).

f. The turkey logs shall be sliced to uniform size, shape, and thickness to ensure compliance with finished product requirements. The turkey slices shall be filled into the tray pack cans within 4 hours after being sliced. End slices shall not be used.

* 3.3.2 Gravy preparation. Gravy shall be formulated and prepared as follows:

Ingredients	Percent by weight
Chicken broth 1/	92.98
Starch 2/	5.50
Onion powder	0.80
Lecithin	0.25
Vegetable oil	0.20
Salt 3/	0.10
Pepper, white, ground	0.05
Celery seed, ground	0.04
Garlic powder	0.03
Sage, ground	0.02
Turmeric, ground	0.01
Marjoram, ground	0.01

to the 5.0 to 5.9 percent solids.

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2/ The total amount of starch in the gravy formula shall be adjusted, as necessary, to produce a product that complies with the finished product viscosity requirements (see 3.6).

3/ The total amount of salt in the gravy formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirement (see 3.6).

NOTE: The following gravy preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared gravy are still applicable.)

The gravy shall be prepared as follows:

a. A slurry shall be made with the starch and part of the chicken broth.

b. The balance of ingredients shall be heated to a boil in a steam jacketed kettle with continuous and vigorous mixing to ensure the spices are uniformly dispersed.

c. The steam supply to the kettle shall be shut off and the slurry added and uniformly mixed to form the gravy.

d. The volume of the gravy shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The gravy shall be maintained in the temperature range of 150° to 180°F and filled into tray pack cans within 4 hours after preparation.

* 3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with turkey slices and gravy to conform to the finished product requirements and to the following requirements:

a. Note less than 54 turkey slices shall be shingled in two rows, lengthwise, in the tray pack can, and then the gravy shall be added.

b. The turkey slices at the time of filling shall be in the temperature range of 28° to 40°F.

c. The gravy at the time of filling shall be in the temperature range of 150° to 180°F.

d. Immediately after filling, each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6w).

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e. The filled and sealed tray pack can shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6.0 has been achieved.

* 3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. No individual can shall contain a bone piece measuring more than 0.3 inch in any dimension.

e. The total weight of skin, cartilage, coarse connective tissue, section of tendons or ligaments, and discolored meat, collectively, in any individual can shall be not more than 1.0 ounce.

f. There shall be no feathers or feather parts.

g. The average net weight shall be not less than 106 ounces.

h. No individual can shall contain less than 104 ounces of product.

i. No individual can shall contain less than 54 slices.

j. Turkey slices shall approximate the size and shape shown in figure 1 or 2.

k. No individual can shall contain more than 3 broken slices.

l. No individual can shall have slices with a void area or air pocket measuring more than 1/2 inch by 1/2 inch.

m. The average drained weight of the slices shall be not less than 63.0 ounces.

n. Drained weight of the 54 turkey slices in any individual can shall be not less than 61.0 ounces.

o. The texture of the slices shall not be dry, rubbery, or mushy.

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p. The turkey slice shall not contain finely ground, flaked, or otherwise comminuted product.

q. The gravy shall be light tan to dark tan in color.

r. The gravy shall not be lumpy.

s. The average fat content shall be not greater than 10.0 percent and no individual tray pack shall have a fat content greater than 12.0 percent.

t. The salt content of any individual tray pack shall be not less than 0.5 percent nor greater than 1.2 percent.

u. The product shall show no evidence of excessive heating (materially darkened or scorched).

v. The viscosity of the gravy shall be not less than 6.0 cm per 10 seconds nor greater than 22.0 cm per 10 seconds when determined by a Bostwick Consistometer.

w. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The turkey component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

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4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

* 4.5.3 Tray pack inspection. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight examination. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty

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tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

* 4.5.3.2 Double sampling plan for product inspection. The finished product shall be examined for the defects listed in table I utilizing the double sampling plans indicated in MIL-STD-105. The lot size shall be expressed in tray pack cans. The sample unit shall be one filled and sealed tray pack can. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for minor defects.

TABLE I. Product defects 1/ 2/ 3/

Category		Defect
Major	Minor	
101		Presence of a bone piece measuring more than 0.3 inch or more in any dimension
102		Presence of feathers or feather parts
103		Total weight of skin, cartilage, coarse connective tissue, section of tendons or ligaments, and discolored meat, collectively, in any individual can is more than 1.0 ounce
104		Less than 54 turkey slices in any individual can
105		Drained weight of 54 turkey slices in a can is less than 61.0 ounces 4/ 5/
106		Slices not approximating the size and shape shown in figure 1 or 2 6/
107		Texture of the slice is dry, rubbery, or mushy
108		Gravy not light tan to dark tan
109		Gravy is lumpy

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TABLE I. Product defects 1/ 2/ 3/ (Cont'd)

Category		Defect
Major	Minor	
110		Product shows evidence of excessive heating (materially darkened or scorched)
111		More than three broken slices
112		More than three slices have a void area or air pocket measuring more than 1/2 inch by 1/2 inch
	201	Three or less broken slices
	202	Three or less slices having a void area or air pocket measuring more than 1/2 inch by 1/2 inch
	203	Slice contains finely ground, flaked, or otherwise comminuted product
	204	Slices not arranged in a shingled manner

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a USDA agent.)

3/ Interior of filled and sealed cans shall be examined visually for coating defects during product examination. Suspected rust spots shall be verified by testing in accordance with 4.5.7. Any spot verified as a rust spot shall be cause for rejection of the lot. Any scratch or fracture that penetrates through the coating of a can shall be cause for rejection of the lot.

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4/ To determine drained weight, the free liquid in the can shall be poured off, strained through a U.S. Standard No. 8 sieve and reserved for viscosity determination. The remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 180° to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat without breaking the turkey slices. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the turkey slices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at such an angle so as to ensure complete drainage of all liquid from the product. The product shall be allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

5/ The lot shall be rejected if the sample average drained weight is less than 63.0 ounces.

6/ Turkey slices, approximating the shape of figure 2, that have broken corners shall not be considered as intact.

* 4.5.3.3 Fat and salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The tray pack cans shall be individually tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, except that preparation of the samples shall be as follows: The unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat and salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

* 4.5.3.4 Viscosity testing. The strained free liquid collected from each of the cans in the first sample of cans selected for drained weight inspection (see 4.5.3.2 and footnote 4/ to table I) shall be individually tested for viscosity as follows (see 6.4).

Instrument: Bostwick Consistometer

Catalog number: 23270-004
VWR Scientific Company
P.O. Box 7900
San Francisco, CA 94120

or

Catalog Number: 15-347-50
Fisher Scientific
585 Alpha Drive
Pittsburgh, PA 15238

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Method:

a. Level the instrument.
b. Bring the gravy to 100°F ± 1°F in a water bath in a covered container.
c. Stir gravy thoroughly before filling the Bostwick cavity.
d. Scrape gravy evenly across upper edge of cavity.
e. Release gravy and time gravy flow to the nearest 1 second and measure distance traveled to the nearest 0.1 cm.

If the Bostwick viscosity value of the gravy from any can in the sample fails to conform to the requirements specified in 3.6, it shall be classified as a major defect and the lot shall be rejected.

* 4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with 4.5.4.1. In addition, the following defect shall be classified as a major defect.

Evidence of buffing with an abrasive substance (see 5.1.1).

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4) except, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can (for example, label raised or peeled back from edges or corners).

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in

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the center of the tray pack can. One measurement shall be made when examining a ribbed lid; lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

* 4.5.7 Test for rust spots on interior of tray pack cans. Where rust spots are suspected on interior of tray pack cans, the following test shall be performed:

a. Immerse a cotton swab in acetone or methyl ethyl ketone solvent and gently rub suspected spot. Handling of the solvent shall be in strict accordance with the guidelines of the manufacturer's Material Safety Data Sheet (MSDS).

b. If suspected spot immediately disappears, it is not a rust spot.

c. If suspected spot remains, continue gently rubbing with the swab resoaked with solvent, and observe for disappearances or retention of spot. Replenish solvent as necessary.

d. If the spot persists, and only a very thin interior can coating, or no interior can coating remains, the spot shall be scored as a rust spot. This can be confirmed further by rubbing with a finger. The rust spot will be felt as a slight bump over the substrate.

4.5.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect or illegible.
Arrangement or number of cans not as specified.

4.5.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

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* 5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5. The practice of reconditioning tray pack cans by buffing with an abrasive substance shall not be permitted (see 4.5.4). The interior coating of filled and sealed thermoprocessed cans shall be free of rust spots, scratches, or fractures that penetrate through the coating when examined in accordance with 4.5.3.2.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

* 5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

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* 5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1990 would be coded as 0078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. In addition, cans shall show the following statements on the lid:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 3 slices each.

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As an alternate labeling method, a preprinted, self-adhering, 0.002-inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack cans shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Level of packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article inspection. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Alternative viscosity testing method. The contracting officer may authorize an alternative contractor-recommended method of viscosity testing if the alternative method is approved by the U.S. Army Natick Research, Development, and Engineering Center.

6.5 Subject term (key word) listing.

Canned foods
Combat field feeding
Operational rations
Shelf stable

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8940-0721)

Review activities:

Army - MD, QM
Navy - MC
DP - SS

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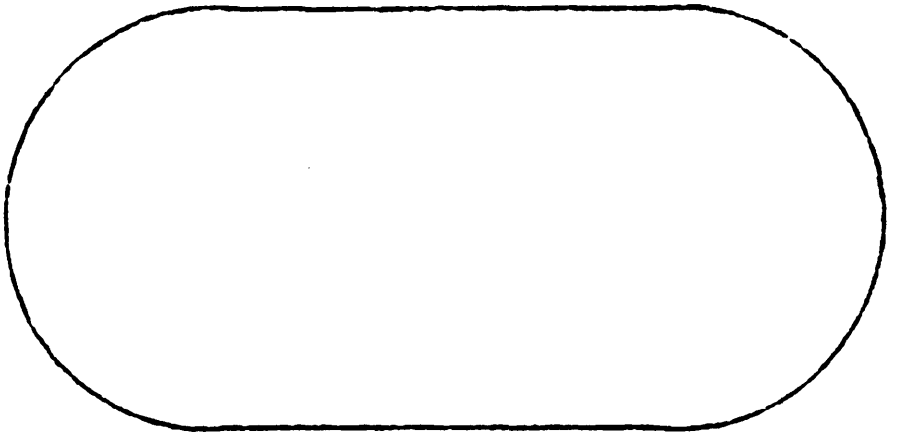


FIGURE 1. Turkey slice shape.

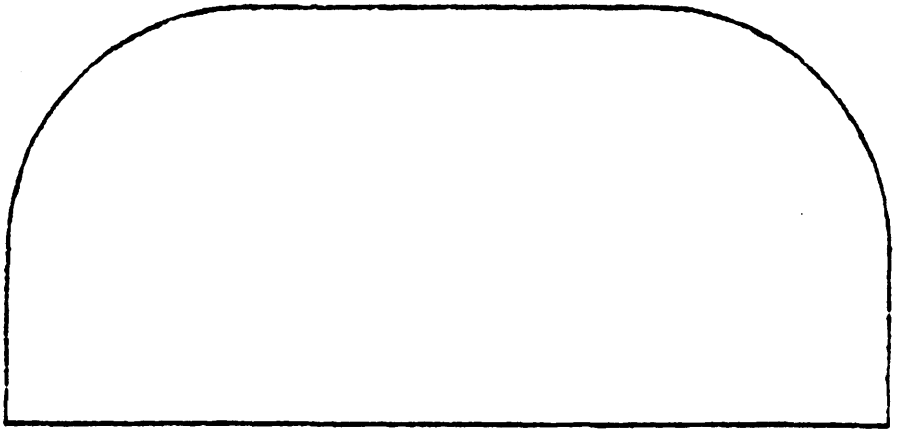


FIGURE 2. Alternate turkey slice shape.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

   The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-44237B	1991 June 30

I. DOCUMENT TITLE

Turkey Slices With Gravy, Thermostabilized, Tray Pack

L. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

. REASON FOR RECOMMENDATION

. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

. PREPARING ACTIVITY

. NAME	b. TELEPHONE (Include Area Code)	
S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON 256-4501
. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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